LINPS: A database for cancer cell-specific perturbations of biological networks

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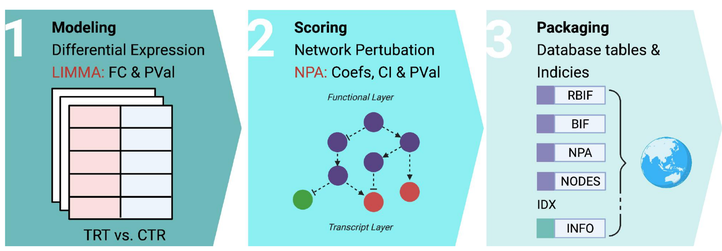
Objectives

- Integrate gene expression of cancer cell lines under drug perutbations and causal biological networks data
- Evaluate the effect of drug treatments on perturbing the biological networks
- Build a database and an interactive interface to make the output of the analysis accessible to all

Summary

Background. Screening for potential cancer therapies using existing large datasets of drug perturbations requires expertise and resources not available to all. This is often a barrier for lab scientists to tap into these valuable resources. To address these issues, one can take advantage of prior knowledge especially those coded in standard formats such as causal biological networks (CBN) [1]. Large datasets can be converted into appropriate structures, analyzed once and the results made freely available in easy-to-use formats. **Methods**. We used the library of integrated library signatures (LINCS) to model the cell-specific effect of hundreds of drug treatments on gene expression [2]. These signatures were then used to predict the effect of the treatments on several causal biological networks using the network perturbation amplitudes (NPA) analysis [4, 3]. **Implementation**. We packaged the pre-computed scores in a database with an interactive web interface. The intuitive user-friendly interface can be used to query the database for drug perturbations and quantify their effect on multiple key biological functions in cancer cell lines. In addition to describing the process of building the database and the interface, we provide a realistic use case to explain how to use and interpret the results. **Conclusion**. To sum, we pre-computed cancer cell-specific perturbation amplitudes of several biological networks and made the output available in a database with an interactive web interface.

Workflow



1. Modeling: the effect of the drug treatments on individual genes in every cell line.
2. Scoring: the fold-change and p-values are used to calculate the network perturbation amplitudes of the drugs on each network (NPA) and each node (Nodes contributions). The sum of the NPA for a group of networks is known as the biological impact factor (BIF).
3. BIF, NPA, and nodes contributions were packaged in a database file and metadata of the cell lines, drugs, and networks.

Contact Information

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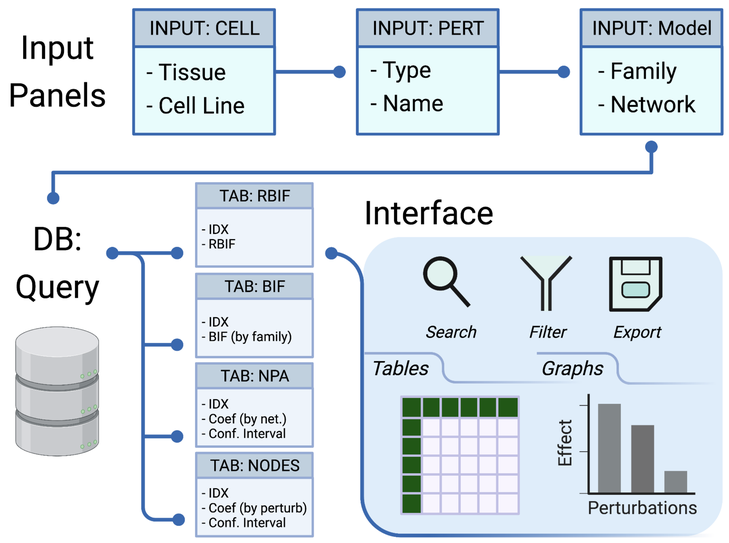
Data Sources

	Cells			Perturbations			Networks		
Desc.	Gene expression profiles of cell lines from different cancer tissue and metadata about the cell phenotypes.			Control vs. treatment gene expression profiles of cancer cell lines and metadata about the treatments.			Two-layers causal biological networks of cellular processes.		
		N	Example		(N)	Example		(N)	Example
Content	Tissue	12	Breast/Skin	MOA	530	NSAID/Anti-ER	Family	5	IPN/CPR
	Cell	70	MCF7/A375	Drug	1938	Asprin/Tamoxifen	Model	8	Apoptosis
Source				LINCS [2]			CBN [1]		

Main Functionality

- Search for drug, network, or cell line of interest
- Filter by most effective perutbation of a family or individual network
- Export output in tabular or graphic formats

Database and interface



- The interface includes input panels to query the database, and the output is sent back to the interface.
- The query is constructed by selecting the cell lines from one or more tissues, the drug perturbations grouped by mechanism of action, and the network models of interest.
- The output of the database query is three main tables: the biological impact factor (BIF), network perturbations amplitudes (NPA), and code contributions (NODES).
- The output is returned in tabular and graphical formats that can be searched, filtered, or exported.

Related work

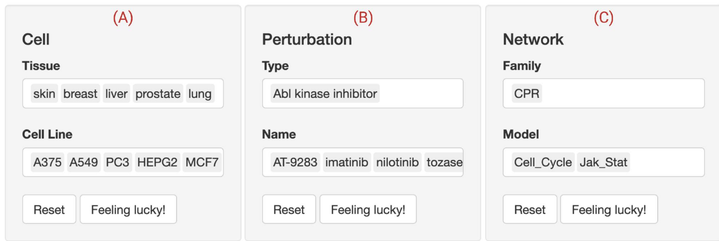
[1] S. Boué et al. "Causal biological network database: A comprehensive platform of causal biological network models focused on the pulmonary and vascular systems". In: *Database* 2015 (2015).

[2] A. Koleti et al. "Data Portal for the Library of Integrated Network-based Cellular Signatures (LINCS) program: Integrated access to diverse large-scale cellular perturbation response data". In: *Nucleic Acids Research* 46.D1 (2018).

[3] F. Martin et al. "NPA: An R package for computing network perturbation amplitudes using gene expression data and two-layer networks". In: *BMC Bioinformatics* 20.1 (2019), pp. 1–9.

[4] F. Martin et al. "Quantification of biological network perturbations for mechanistic insight and diagnostics using two-layer causal models". In: *BMC Bioinformatics* 15.1 (2014).

The input panel

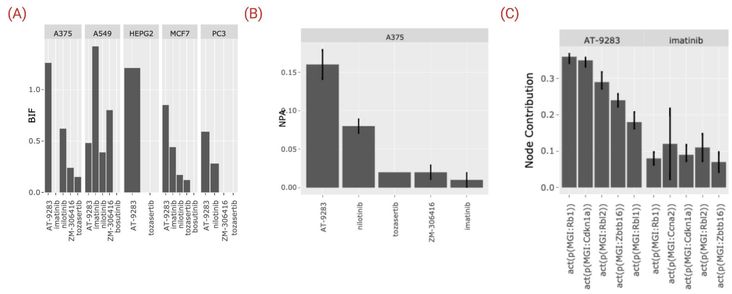


(A) Cell, is to choose the tissue and cellular context where the drug perturbations would be shown.

(B) Perturbation, allows for searching and selecting the type (mechanism of action) and name of the drugs of interest.

(C) Network, lists the available biological networks to be scored classified into multiple families.

The graphical output

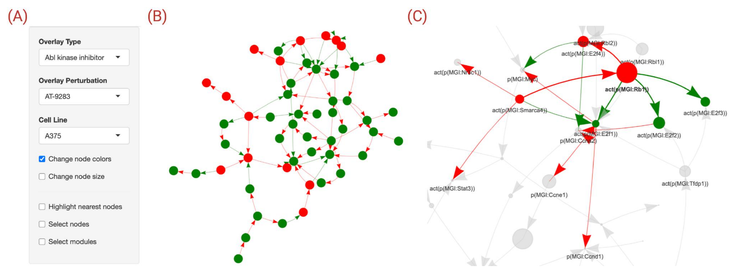


(A) Biological impact factor (BIF) of every drug on the cell proliferation (CPR) networks stratified by the cell line.

(B) Network perturbation amplitudes (NPA) of every drug on the "cell cycle" biological network in A375.

(C) The relative contribution of each node (top five) in the cell cycle network in response to the drug perturbations in A375.

The graph representation



(A) Input panel to search and select from the main inputs for individual perturbations to visualize on the network graph.

(B) The "cell cycle" network shown as a graph overlaid with the effect of "AT-9283" in the "A375" cell line.

(C) Nodes are colored by the direction of regulation in response to the drug treatment. Edges are colored by regulation direction between the nodes.

Scan the QR code for more.